Top 100 Orthopedic AI • Rank #64

AI-augmented personalized treatment in healthcare a comprehensive survey of models benchmarking and deployment challenges

Vijayalaxmi N. Rathod, R. H. Goudar, Geetabai S. Hukkeri

Journal:	Publication Year:	Citations:	DOI:
Discover Artificial Intelligence	2026.0	0	10.1007/s44163-026-01187-2

CLINICAL & AGENTIC AI TAKEAWAY

This survey highlights the need for robust, transparent, and clinically viable personalized treatment recommendation systems in healthcare, addressing challenges in model interpretability, population generalizability, data privacy, and ethical compliance.

ABSTRACT

Personalized Treatment Recommendation Systems (PTRS) are increasingly central to precision medicine, enabling data-driven selection of therapies tailored to individual patient characteristics, including clinical history, biological markers, and behavioral context. Despite rapid methodological advances, existing research remains fragmented, with limited consensus on evaluation practices and insufficient consideration of real-world deployment constraints. This survey provides a systematic and critical review of 60 peer-reviewed studies published between 2019 and 2024, selected using a PRISMA guided protocol from major scholarly databases. The reviewed literature is organized into a unified taxonomy encompassing content-based, collaborative filtering, knowledge-based, and hybrid treatment recommendation paradigms, with emphasis on their clinical applicability and algorithmic design. Recent advances in deep learning and reinforcement learning are examined for their role in improving adaptability and treatment personalization. To enable meaningful cross-study comparison, this work introduces a multidimensional benchmarking rubric that evaluates PTRS beyond predictive performance, incorporating criteria such as algorithmic rigor, clinical relevance, validation depth, data transparency, scalability, and explainability. The analysis reveals persistent challenges related to model interpretability, population generalizability, data privacy, and ethical compliance, which continue to hinder clinical adoption. Emerging solutions, including federated learning, explainable AI, and privacy-preserving architectures, are discussed as promising pathways toward trustworthy deployment. By synthesizing methodological trends, practical limitations, and evaluation gaps, this survey offers a structured foundation and forward-looking perspective for the development of robust, transparent, and clinically viable PTRS in modern healthcare.

Study Overview & Background

Personalized Treatment Recommendation Systems (PTRS) are increasingly central to precision medicine, enabling data-driven selection of therapies tailored to individual patient characteristics, including clinical history, biological markers, and behavioral context. Despite rapid methodological advances, existing research remains fragmented, with limited consensus on evaluation practices and insufficient consideration of real-world deployment constraints. This survey provides a systematic and critical review of 60 peer-reviewed studies published between 2019 and 2024, selected using a PRISMA

guided protocol from major scholarly databases. The reviewed literature is organized into a unified taxonomy encompassing content-based, collaborative filtering, knowledge-based, and hybrid treatment recommendation paradigms, with emphasis on their clinical applicability and algorithmic design. Recent advances in deep learning and reinforcement learning are examined for their role in improving adaptability and treatment personalization. To enable meaningful cross-study comparison, this work introduces a multidimensional benchmarking rubric that evaluates PTRS beyond predictive performance, incorporating criteria such as algorithmic rigor, clinical relevance, validation depth, data transparency, scalability, and explainability. The analysis reveals persistent challenges related to model interpretability, population generalizability, data privacy, and ethical compliance, which continue to hinder clinical adoption. Emerging solutions, including federated learning, explainable AI, and privacy-preserving architectures, are discussed as promising pathways toward trustworthy deployment. By synthesizing methodological trends, practical limitations, and evaluation gaps, this survey offers a structured foundation and forward-looking perspective for the development of robust, transparent, and clinically viable PTRS in modern healthcare.

Methodology & Clinical Focus

Study Classification: systematic review. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

This survey highlights the need for robust, transparent, and clinically viable personalized treatment recommendation systems in healthcare, addressing challenges in model interpretability, population generalizability, data privacy, and ethical compliance.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

- **Authors:** Vijayalaxmi N. Rathod, R. H. Goudar, Geetabai S. Hukkeri
- **Journal / Venue:** Discover Artificial Intelligence (2026.0)
- **SJR Metric:** Quartile 1.0
- **Citation Count:** 0 citations recorded
- **Digital Object Identifier (DOI):** https://doi.org/10.1007/s44163-026-01187-2
- **Consensus Evidence Link:** https://consensus.app/papers/aiaugmented-personalized-treatment-in-healthcare-a-rathod-goudar/ef7219f92bd05820bee86bc95ec9d800/

Source & Provenance: Curated for Agentic Orthopedics Research Collection • DOI: https://doi.org/10.1007/s44163-026-01187-2 • Consensus Page: https://consensus.app/papers/aiaugmented-personalized-treatment-in-healthcare-a-rathod-goudar/ef7219f92bd05820bee86bc95ec9d800/ • Verified and rendered via Deep Search & Academic Repositories.